

Monotone Subsequences

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to create a permutation of numbers $1, 2, \dots, n$ whose longest monotone subsequence has exactly k elements.

A monotone subsequence is either increasing or decreasing. For example, some monotone subsequences in $[2, 1, 4, 5, 3]$ are $[2, 4, 5]$ and $[4, 3]$.

Input

The first input line has an integer t : the number of tests.

After this, there are t lines. Each line has two integers n and k .

Output

For each test, print a line that contains the permutation. You can print any valid solution. If there are no solutions, print IMPOSSIBLE.

Constraints

- $1 \leq t \leq 1000$
- $1 \leq k \leq n \leq 100$

Example

Input	Output
3 5 3 5 2 7 7	2 1 4 5 3 IMPOSSIBLE 1 2 3 4 5 6 7